Process Design Proposal to Synthesize 25,000 tonne/year Dimethylamine Through Methylamine Synthesis in Aspen Plus

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Abstract

In aiming to create a process design that produced 25,000 tonnes/year Dimethylamine, we formulated a design schematic with a reactor and multiple columns in series in order to separate the products of our reactor. First, an an equilibrium reactor was modeled that reacted Ammonia and Methanol to create an equilibrium mixture of Methylamine (MMA), Dimethylamine (DMA), and Trimethylamine (TMA), followed by 3 distillation columns to remove excess ammonia from the reactor outlet. From there the bottoms streams were sent with excess water to a RadFrac extractive distillation column to remove and recycle TMA, the bottoms stream to a flash drum to remove excess water, and a final distillation column to separate the MMA and DMA products. Based on the initial analysis, it has been found that an equal-weight feed of 130,000 tonnes/year each of Ammonia and Methanol produced approximately the target amount of DMA, economic analysis of potential profits included a Gross Margin of 6031 MM$/oper-yr and an NPV of 9597 MM$ after an operation of 10 years.

Economic analysis

This design provides an annual gross margin of 6031 MM$/oper-year from a revenue of 38966 MM$ and raw material cost of 32934 MM$. Using installation costs and sizing parameters, a fixed cost of 1927 MM$ is achieved. However, the separation columns energy consumption takes up a majority of the operating costs.

Component	Value (tonne/oper-year)	Price ($/oper-year)
Ammonia	58,500	24,255,855,000
Methanol	146250	8,678,796,750
	Total Raw Material Cost	**32,934,651,750**
Ammonia	33,490	13,885,958,700
MMA	11,159	597,921,538
DMA	25,617	9,990,630,000
TMA	12,105	12,105,000,000
	Total Revenue	**38,966,159,336**
	GROSS MARGIN	**6,031,507,586**

Fig. 1, final process design materials costs and revenue from products

Motivations

Our main goal was to produce 25,000 tonnes/yr DMA, which is the methylamine that's in the most demand worldwide using a series of separation columns. All 3 methylamines (MMA, DMA, and TMA) all have uses in industry, primarily as intermediate chemicals and additives, however the demand between the three is between 1:3:1 and 1:8:1 in contrast to an equilibrium ratio of 5:6:9. To maximise our production of DMA, we optimized our reactor with catalysts that favored DMA production and added recycle streams to maximise our yield and reduce raw materials costs

Methods

We used Aspen Plus process simulation software to simulate our process and optimize our unit operations. We chose to use the *Amines* property method and based our initial layout on that commonly found in literature, namely a reactor followed by an ammonia separating step, a TMA extraction step, a dehydration step, and a DMA/MMA separation step.

We found that only using one column to remove ammonia from our reactor products wasn't sufficient so we expanded to 3 columns. The TMA extraction column was chosen to be a RADFRAC column to account for the azeotropic nature of the methylamines mixture with another column added to ensure the purity of our TMA byproduct. A flash drum was found to be sufficient for dehydrating our products and a standard DSTWU column with a secondary flash drum added to purify the MMA byproduct and recycle raw components. Recycle streams were added to lower the feed rate of fresh materials and lower overall operating costs

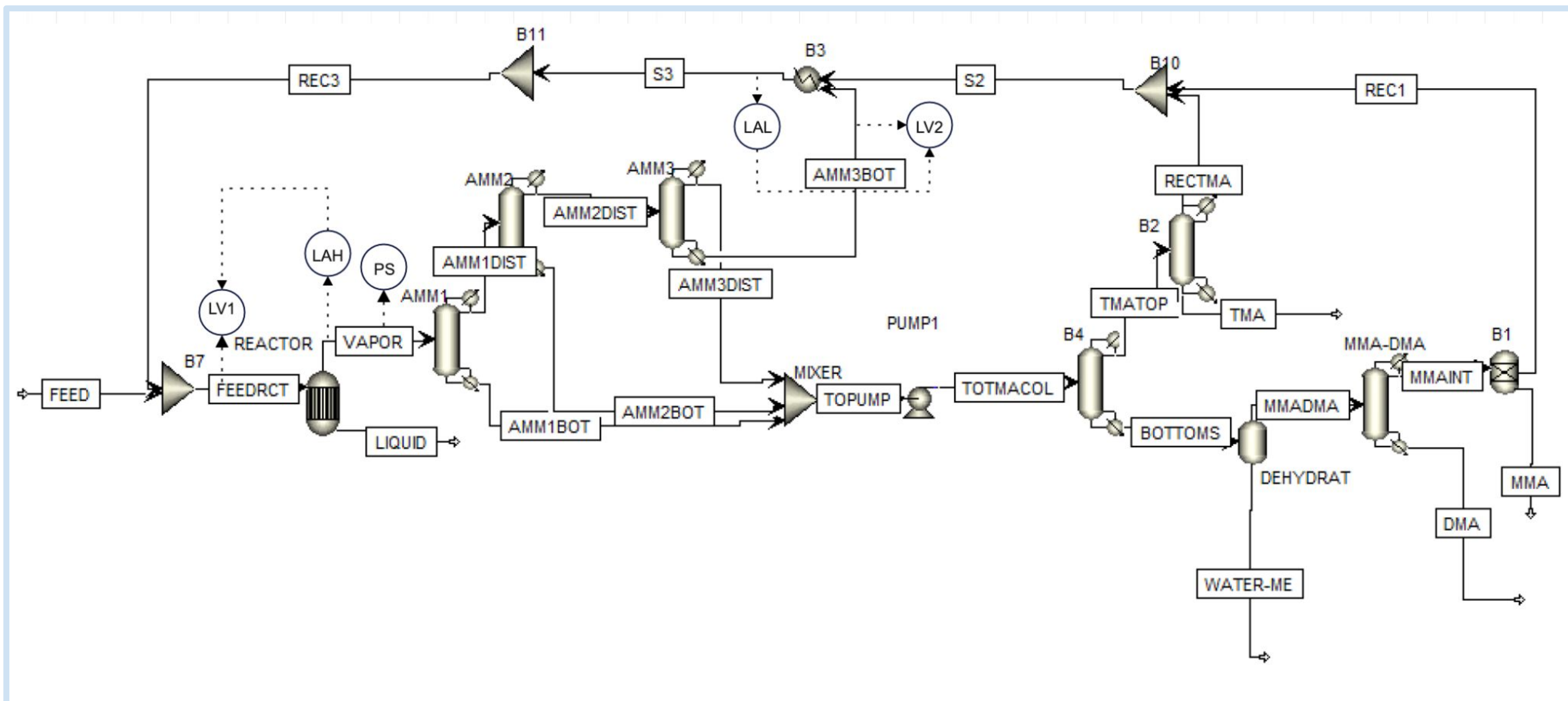

Fig. 2, P&ID diagram of the final process design, stream tables can be found by following the QR code in the top left corner

Design Results

Unit Operation	Energy Requirement (Btu/hr)
Reactor	1.10E+07
Ammonia 1	3.0E+07
Ammonia 2	3.9E+07
Ammonia 3	4.6E+07
TMA (B3) Condenser	5.65E+06
TMA (B3) Reboiler	2.44E+07
TMA Separator	7,060.35
Dehydration Flash	2.68E+07
MMA/DMA	1.65E+06
MMA/DMA Separator	1.52E+06
Total	**1.87E+08**

Fig. 4, Energy requirements per unit operation

Parameter	Reactor	Ammonia 1	Ammonia 2	Ammonia 3
Diameter	7.5	9	5	8
Height	30	40	20	32
Tray Spacing		2	2	2

Parameter	Dehydration Flash	TMA	TMA Separator	MMA/DMA	MMA/DMA Separator
Diameter	4	10	3.5	6	3
Height	12	22	12	14	12
Tray Spacing		2		2	

Fig. 5, final main equipment sizing information

Energy Requirement

Component	Reactant (tonne/oper-year)	Product (tonne/oper-year)
Ammonia	115,000	42,272.27
Methanol	115,000	601.64
MMA	0	38,553.56
DMA	0	7,439.02
TMA	0	669.62
Water	0	5,521.94
	Energy Required	**46,645,597.5 Btu/hr**

Component	Reactant (tonne/oper-year)	Product (tonne/oper-year)
Ammonia	58,500	8,211.03
Methanol	146,250	2,835.29
MMA	0	33,459.67
DMA	0	25,617.88
TMA	0	9,245.14
Water	0	3,229.12
	Energy Required	**11,030,629.3 Btu/hr**

Fig. 3a-b, final process design energy requirements without any recycle streams (a) and with the recycle stream present in Fig. 2 (b)

Conclusions

By using guidance from literature and formulating a process that uses multiple ammonium extraction columns, a TMA extraction column and recycling step, and various unit operations to separate and purify our DMA and MMA products, we were able to make a process that made and purified 25,000 tonne/oper-year of DMA, as per our project guidelines. The annual operating cost of our final design was 973 MM$/yr with an NPV value after 10 years of 9597 MM$. The gross margin was also found to be 6031 MM$/oper-yr. Adding a recycle stream certainly helped in bringing down the raw materials cost of this project, as without the recycle stream it came out to 32934 mm$/oper-yr. Future ar29 eas of work in this project includes implementing heat exchangers, particularly before certain unit operations such as the reactor and the TMA column, and furthur optimizing the recycle stream, as in its current state it increases the temperature of several unit operations (such as the TMA column and a couple of the ammonium extraction columns) that cause errors and non-ideal results.